Rushendra Sidibomma

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Education

MS Computer Science, University of Minnesota Twin Cities (**GPA: 3.94/4.0**) **May 2027 (Expected)**
Coursework: Computer Vision, Behavioral Data Mining, Natural Language Processing, Deep Learning

BS Computer Science, IIIT Sri City (**GPA: 3.82/4.0**) **2020 - 2024**
Coursework: Data Structures, Algorithms, Database Systems, Computer Networks, Data Mining, Cloud Computing

Technical Skills

Languages: Python, Go, SQL, JavaScript, Java, Bash
LLMs: OCR, Strcutured Extraction, Document AI, Vision-Language Models (VLMs), RAG, LLM Evaluation, Agent Orchestration, Tool Use
ML: PyTorch, Transformers, Hugging Face, Scikit-Learn, RAPIDS, MLOps, Computer Vision, RLHF/RLVR, Fine-tuning, LoRA
Backend / APIs: Django, FastAPI, Node.js, REST APIs, API Design, Async Workers, Microservices
Databases / Infrastructure: PostgreSQL, MySQL, MongoDB, Redis, AWS, Azure, Docker, CI/CD, Linux, Git

Industry Experience

BioStack Platforms (YC P26) **May 2026 – Present**
AI Development Intern *San Francisco, CA*

- Designed and implemented 15+ reinforcement-learning environments for LLM agents, converting longitudinal clinical workflows into measurable states, actions, rewards, and task success criteria for post-training.
- Generated 300+ LLM-agent scenarios across reasoning, retrieval, and tool-use workflows, using structured traces and scoring rubrics to support RLHF/RLVR data generation and failure analysis.
- Improved LLM-agent task success rate by 18% and reduced unsupported responses by 25% by converting failure traces into targeted red-teaming cases, reward refinements, and post-training data signals.

AutomationEdge Technologies **July 2024 – July 2025**
Machine Learning Engineer *Pune, India*

- Built a production document-intelligence platform combining OCR, VLM-based layout handling, agentic LLM extraction, and schema validation to convert unstructured clinical documents into validated records; reduced patient intake errors by 40%.
- Fine-tuned a VLM-based layout classifier to detect multi-column layouts, checkboxes, and other complex page formats, routing only high-value pages to frontier vision models and reducing vision-model inference costs by over 50%.
- Devised keyword-based page pruning to skip 60% of irrelevant pages before extraction, cutting model-processing costs by 40% while preserving field-level recall on gold-standard documents.
- Implemented Django REST APIs and PostgreSQL-backed workflow state for upload, parsing, extraction, validation, audit logs, and RPA handoffs; kept large-document latency under 90 seconds across 20,000+ monthly requests.
- Built a pytest-based regression suite over gold-standard medical documents to validate schema correctness and field-level extraction quality despite nondeterministic LLM outputs; kept extraction error rate under 10%.

Eyemote Vision **Sep 2023 – Jan 2024**
Backend Engineering Intern *Remote*

- Engineered a backend ML inference service in Python for detecting diabetic retinopathy from retinal fundus images, exposing deep learning model predictions via APIs for integration into clinical screening workflows.
- Trained and deployed computer vision models in PyTorch to grade severity of diabetic retinopathy on large-scale retinal imaging datasets, optimizing inference for low-latency, near real-time predictions.
- Integrated the ML system with partner eye clinics' internal systems to enable automated image ingestion and delivery of diagnostic predictions in production healthcare environments, shipping the system as the minimum viable product (MVP).

PROJECTS

LLM Agent for Enterprise Ticket Routing | *Django, CrewAI, Fine-tuned LLMs, REST APIs*

- Built a CrewAI-based LLM agent for ticket classification and routing, using a fine-tuned distilled model that achieved 93% accuracy at 70% lower inference cost than frontier-model baselines.
- Combined LLM reasoning, rule-based validation, and REST API integrations to assign tickets, trigger resolution workflows, and guard against invalid actions.
- Deployed as an internal project for enterprise IT operations, processing 500+ daily requests and reducing manual triage effort.

RAG-Based Knowledge Management System | *Django, RAG, OpenAI API, AWS Bedrock, Azure AI Search, AWS OpenSearch*

- Built document ingestion and retrieval pipelines for enterprise policy content, supporting configurable chunking, embeddings, vector stores, and LLM generation across AWS and Azure cloud infrastructure.
- Designed provider-agnostic RAG workflows across Azure AI Search and AWS OpenSearch, with validation guardrails to reduce unsupported responses and improve retrieval reliability.

Risk-Adjusted Hallucination Detection for Grounded QA | *Python, scikit-learn, LLM Evaluation, Calibration, Selective Prediction*

- Developed a hallucination-risk detection pipeline for grounded QA using uncertainty, self-consistency, semantic entropy, and evidence-consistency features; trained calibrated logistic regression detectors across PHANTOM and WikiQA.
- Used calibrated risk scores for selective prediction and abstention, achieving 0.773 AUROC / 0.645 AUPRC on PHANTOM and improving selective accuracy from 75.9% to 78.6% by abstaining on the highest-risk 18% of cases.

Research Experience

Masonic Cancer Research Center, University of Minnesota — **January 2026 – May 2026**
Graduate Research Assistant — *Minneapolis, MN*

- Engineered Python pipelines for 230GB+ high-resolution microscopy images, handling preprocessing, segmentation, and structured feature extraction across diverse multiplex histology datasets.
- Fine-tuned deep learning-based segmentation models using PyTorch to extract features from 1M+ cells. Improved the segmentation accuracy of standard Cellpose models by 10% on internal lab-generated imaging datasets.
- Developed GPU-accelerated clustering and analysis workflows with RAPIDS, Pandas, and NumPy, reducing manual review time by 60% and improving cell identification and classification reliability.

Trustworthy AI Lab, Toronto Metropolitan University — **May 2023 – May 2025**
*Mitacs Fellow – **Received a fellowship of $10,000, acceptance rate of <8% globally among 25,000 applicants*** — *Toronto, Canada*

- Built scalable verification frameworks to evaluate neural network reliability under adversarial conditions, reducing certification runtime by 70% compared to prior methods.
- Designed optimization-based certification pipelines that improved computational efficiency by 47% while maintaining less than 2% deviation from comparable high-cost state-of-the-art methods.
- Developed reproducible benchmarking methods to compare model robustness across architectures, supporting systematic model validation and safety assessment of deep learning models.

Research Publications

- Maleki, **Sidibomma** et al. Cascading Robustness Verification. [*IEEE SaTML 2026*] [Link]
- **Sidibomma** et al. Hate Speech Detection in Devanagari Languages. [*CHiPSAL @ COLING, 2025*] [Link]
- **Sidibomma**, Sanodiya. Semantic Representations for Unsupervised Domain Adaptation. [*IEEE ESDC, 2023*][Link]
- Maleki, **Sidibomma** et al. Certified Robustness at Scale via Sparse SOCP Relaxations. [*Under review*]